

METU Computer Engineering

CEng 111 – Fall 2026
Week 1a

Course Overview &
Introduction to CS

Credit: Some slides are from the “Invitation to Computer Science” book by G. M. Schneider, J. L. Gersting and some from the “Digital Design” book by M. M. Mano and M. D. Ciletti.



Today

- Course Overview
  - Syllabus
  - What the course is about
  - Homeworks, Labs
  - The book
- Introduction to Computer Science
  - What CS is and what it is not
  - Computing, computation, algorithm



COURSE OVERVIEW

Syllabus

https://ceng.metu.edu.tr/~skalkan/ceng111_syllabus.pdf

CENG 111
Introduction to Computer Engineering Concepts
2026-2027 Fall

Instructors:
Dr. Hüseyin Aydın, Dr. Sinan Kalkan

Teaching Assistants:
Orhun Buğra Baran, Defne Erkan, Dursun Ece Gökçay, Ersel Hengirmen, Defne Odabaşı, Umut Özdemir, Yiğit Sever, Selay Tekgül, Bora Yalçıner

Course Schedule:

Section 1	Section 2
Tue: 11:40, 12:40 (BMB-1) Wed: 11:40 (BMB-1)	Tue: 9:40, 10:40 (BMB-1) Thu: 10:40 (BMB-1)

Lab Schedule: TBA. **Prerequisites:** None

Contact

- Office hours: By appointment.
- Hüseyin Aydın:
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  - Email: huseyin@ceng.metu.edu.tr
  - Web: https://user.ceng.metu.edu.tr/~huseyin/
- Sinan Kalkan:
  - Office: A-209
  - Email: skalkan@metu.edu.tr
  - Web: www.ceng.metu.edu.tr/~skalkan
  - Twitter: www.twitter.com/kalkansinan



Labs

- There will be labs almost every week
- Labs are complementary to the lectures
  - Lectures are mostly theoretical
  - Labs give you the practical knowledge/abilities
  - Attendance is mandatory: Skipping two lab sessions will result in NA grade.
- Take them extremely seriously!

Content

METU Computer Engineering

Catalog Description:

Introduction to the fundamentals of computer systems, including computer organization, operating systems, language processors and user interfaces.Introduction to algorithms and programming. Reasoning informally about the correctness and efficiency of programs. A functional programming language will be used for practical work.

- First part of the course:
  - What is CS, computing, computation, algorithm?
  - Inside of a Computer: CPU, Memory, Gates, Boolean Logic.
  - Von Neumann architecture
  - Etc.
- Second part of the course:
  - Programming concepts with Python.

Text Books

- *For the 1. part:* "Introduction to Computer Hardware and Systems: Lecture Notes", by G. Üçoluk and S. Kalkan, 2024.
  - *Uploaded to the ODTUclass page of the course.*

- *For the 2. part:* "Introduction to Programming Concepts with Case Studies in Python", by G. Üçoluk and S. Kalkan, Springer, 2012.
  - *https://link.springer.com/book/10.1007/978-3-7091-1343-1*
  - *Downloadable from METU*
  - *Uploaded to the ODTUclass page of the course.*

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The textbook

AVAILABLE AT THE PHOTOCOPY ROOM AT THE BASEMENT OF BUILDING-A (DEREAGZI FOTOKOPI).

WE DO NOT EARN ANY MONEY FROM THIS!



Göktürk Üçoluk

Göktürk Üçoluk received his B.Sc. and M.Sc. degrees in Physics from the Bosphorus University, İstanbul, Turkey, and his Ph.D. degree from the Middle East Technical University, Ankara, Turkey, in 1980, 1982, and 1989, respectively. He is the developer of several symbolic computation software, among which are the first freeware Standard Lisp interpreter and the cross compiler.

He is currently a full professor at the Department of Computer Engineering at the Middle East Technical University. His recent interests are in evolutionary computing, modeling and simulation. His other research interests include symbolic algebraic computing, language processors, and natural language processing. Along with lectures in his subject field, he also has taught over 20 years the departmental freshman CS courses.



Sinan Kalkan

Sinan Kalkan graduated from the Middle East Technical University with a B.Sc. degree and a M.Sc. degree in Computer Engineering in 2001 and 2003, respectively. He received his Ph.D. degree in Informatics from one of the 5 elite universities of Germany, the University of Göttingen, in 2008.

He is currently an assistant professor at the Department of Computer Engineering at the Middle East Technical University, teaching freshman and sophomore courses as well as graduate level advanced courses. His research interests include Pattern Recognition, Image Processing, Computer Vision and Cognitive and Swarm Robotics. He is one of the main contributors of an open source computer vision library, CoViS (http://www.covig.org).



Üçoluk
Kalkan

INTRODUCTION TO PROGRAMMING CONCEPTS



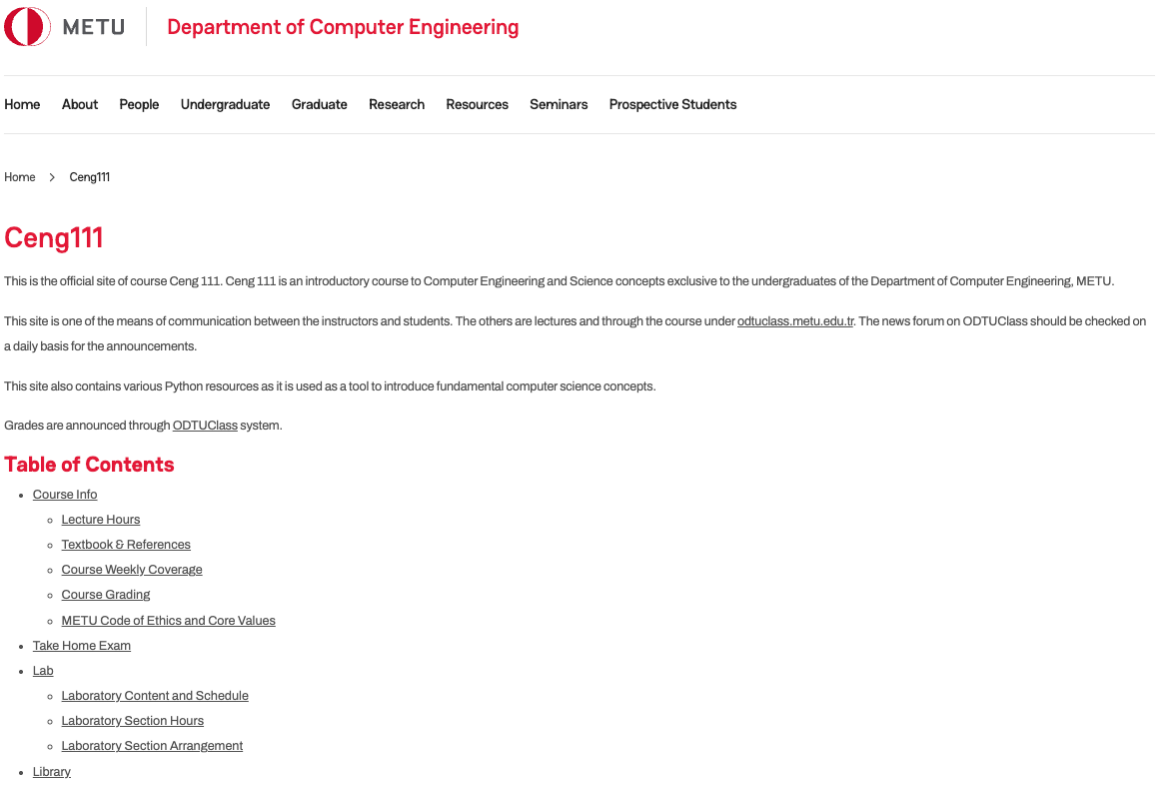
INTRODUCTION TO PROGRAMMING CONCEPTS

with case STUDIES in Python

Göktürk Üçoluk Sinan Kalkan

Home page

- http://ceng.metu.edu.tr/ceng111



METU | Department of Computer Engineering

Home About People Undergraduate Graduate Research Resources Seminars Prospective Students

Home > Ceng111

Ceng111

This is the official site of course Ceng 111. Ceng 111 is an introductory course to Computer Engineering and Science concepts exclusive to the undergraduates of the Department of Computer Engineering, METU.

This site is one of the means of communication between the instructors and students. The others are lectures and through the course under odtuclass.metu.edu.tr. The news forum on ODTUClass should be checked on a daily basis for the announcements.

This site also contains various Python resources as it is used as a tool to introduce fundamental computer science concepts.

Grades are announced through ODTUClass system.

Table of Contents

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  - Laboratory Section Hours
  - Laboratory Section Arrangement
- Library



CENG account

- user name/passwd will be mailed to your METU account (the one **used for registration**).
- **use**

webmail.metu.edu.tr

to view mails there.

Grading

Grading:

Midterm	%27
Lab Exams (3-4)	%30
Von Neumann Machine Exam	%5
Participation	%3
Final	%35

- Most assignments will be programming assignments (some could be exercises to be written up).
- Laboratory schedule will be announced later. Keep watching.
- Starting with the Weeks 4-5 the laboratory sessions will go in parallel with the lectures. You will be asked to solve problems by using the ideas introduced in the lectures of that week. Earlier lab sessions will build your computing/communicating literacy.
- Attending the lab sessions is mandatory. Missing two lab sessions will result in a grade of NA for the course.



- Attending the lab sessions is mandatory. Missing two lab sessions will result in a grade of NA for the course.



Take-home Exams

- We will have four take-home exams:
  - Heavy programming!
- They are extremely important:
  - You get the chance to practice what you learn
  - You can get your hands “dirty”
  - Well, they are your first “few walking steps” towards programming

No Take-home Exams anymore!



Lab Exams

- We will have around four lab exams
- One exam on Linux basics
- The remaining on programming with Python
  - We will share a sample problem 2-3 weeks before the exam.
  - You are expected to solve the problem and submit your solution to us before the exam.
  - There will be a programming exam on a similar problem, in a computer lab without internet access.



Lab Exams

- They are extremely important:
  - You get the chance to practice what you learn
  - You can get your hands “dirty”
  - Well, they are your first “few walking steps” towards programming

The use of LLMs (AI models) & Lab Exams

- In this department, we teach you a new occupation!
  - This means you have to be an expert on your own in designing solutions (algorithms) to problems and coding your solutions with a programming language
- Yes, you will use AI models for coding after your graduation
- However, if you rely on an AI model for understanding and writing code during your courses, you will not be able to develop and hone your engineering skills

Cheating & Lab Exams

- In this department, we teach you a new occupation!
  - In this sense, if you cheat, you fool yourselves; not us!
- Cheating is not acceptable in this department or any other departments or teaching institutions.
- Violation => disciplinary action!!



- The following are not accepted as excuses for cheating (in fact, nothing is acceptable for cheating):
  - “Ama hocam, ödev metninde yazmamışsınız kopya çekemeyeceğimizi”
  - “Ama hocam, üst sınıflar «çekin çekin, bi şey olmuyo, ben geçtim» demişti”
  - “Ama hocam, arkadaşın kodunu değiştirdim, aynısını göndermedim ki!”
  - “Ama hocam, bir tek ben çekmedim, herkes birbirine mail atıyor kodunu”

Academic Integrity
What is it?
Why does it matter?

The Clemson University International Center for Academic Integrity (October, 1999).

Borrowed from Prof. Fatos Yarman Vural

Integrity

Doing the right thing, even, when no one is watching!

Clive Staples Lewis

Borrowed from Prof. Fatos Yarman Vural

Academic integrity

<<"Academic integrity" is the **honesty in all matters** relating to endeavor of the academic environment>>

- Teaching and learning of knowledge and skills,
- Discovery and dissemination of new knowledge.

Turner & Beemsterboer (2003)

Borrowed from Prof. Fatos Yarman Vural

Honesty to Oneself and Others

- Courage and insight to make difficult choices,
- Responsibility for actions and their consequences, even at personal cost,
- Fair treatment among all students and faculty, **regardless of any condition,**
- Personal responsibility to assist in preventing academic dishonesty.

Borrowed from Prof. Fatos Yarman Vural

Academic Dishonesty

- Copying and/or purchasing from **another student** or **internet sources**,
- Allowing another student to share solution during an exam,
- Faking the results of a computer program,
- Using material from another source without credit to the owner,
- Working in a group, when an assignment was for individual work.

Borrowed from Prof. Fatos Yarman Vural

Academic Dishonesty

- Ignoring the balance in the workload in team work assignments and hiding it from the instructor
- Attempting to cheat the members in teamwork and act as if successful

Ariely (2012), Adams (1990)

Borrowed from Prof. Fatos Yarman Vural

Using Web and AI Tools

- There is a distinction between
  - helping or receiving help from some tool or your classmates and
  - cheating or stealing.

- You are responsible for understanding the difference between
  - using the web/AI for help and support of your learning, versus using the web/AI to cheat.
  - communicating with your classmates in an honor system, versus lying, cheating or stealing.

Borrowed from Prof. Fatos Yarman Vural

How to Avoid Academic Dishonesty?

- Improve your abilities for intrinsic motivation
- Always keep in mind:
  - Getting a poor grade is better than pretending as if you were successful.
  - Losing status among peers is better then being a cheater.
  - Observing cheaters does not make you eligible to cheat.
  - **Although you may win in the short term, you will be a loser in your life, if you cheat!**

Borrowed from Prof. Fatos Yarman Vural

Epilog

If you want inner peace, you'll be honest. Because; cheating always condemns you to live on thorns.

- Celaleddin Rumi

If it is not right, do not do it; if it is not true, do not say it.

- Stoic Rule

Borrowed from Prof. Fatos Yarman Vural

Motivation

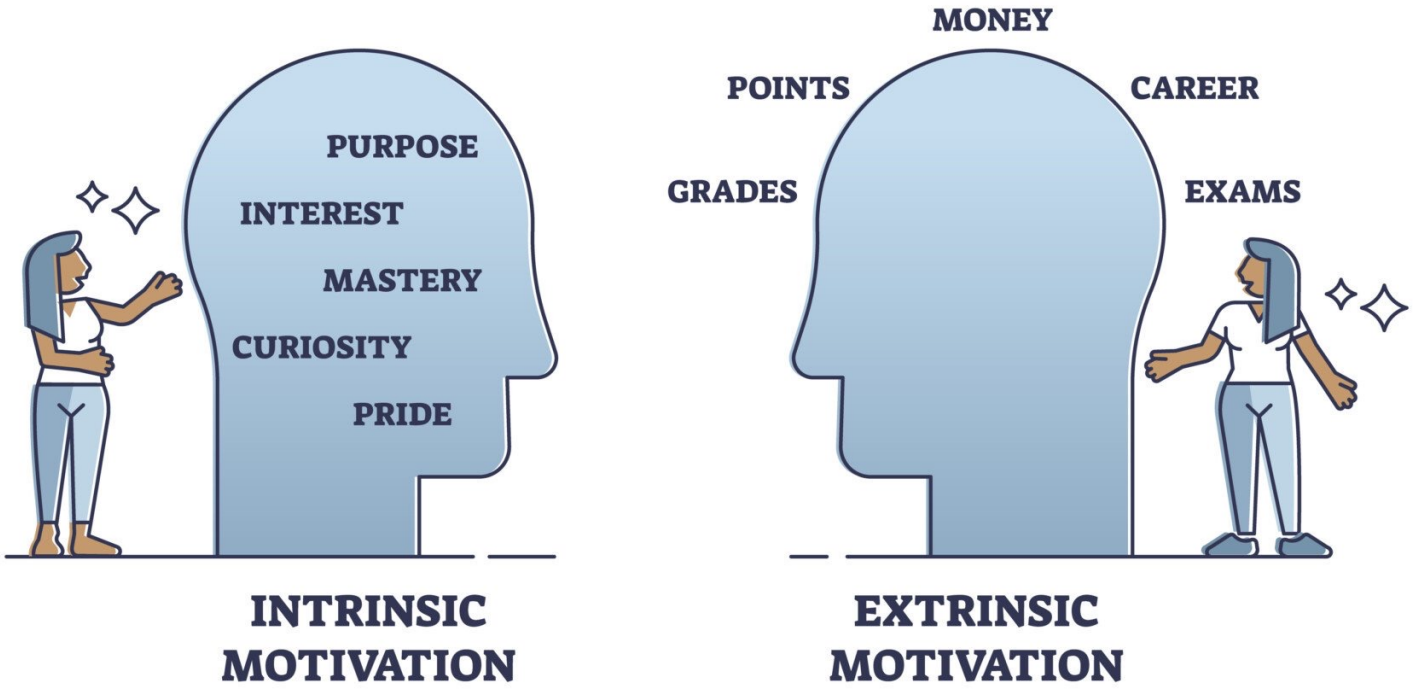


Figure: https://www.simplypsychology.org/differences-between-extrinsic-and-intrinsic-motivation.html



Motivation

- Watch a video by Dr. Huberman on the biological basis for motivation & perspective:

  https://www.youtube.com/watch?v=QmOF0crdyRU



COMMON QUESTIONS ABOUT THE COURSE & THE DISCIPLINE

Common questions about CENG

- Will AI eliminate our jobs?
- No, because:
  - AI will always make errors (hallucinate, miss the context etc.)
  - AI will not be trustworthy
  - AI can be malicious
  - There will always be human experts using AI as a tool (human in the loop systems)
- For more information, you can watch the following video:

"Artificial Intelligence and the Wizard of Oz" (Sinan Kalkan)
https://www.youtube.com/watch?v=MdN7sDrzWNE



Common questions about CENG

- Will AI eliminate our jobs?
- Will we write code in professional life?
- How will we find a job?
- An academic vs. professional path?
- How shall I improve myself in my free time?
- Do CENG people have a “Chamber”?

Nature of CENG111

- How is the course going to be conducted?

- How to study?
  - Take detailed notes by hand-writing during lectures (why? It helps learning & internalizing new knowledge [1])
  - Go over your notes before the day ends
    - Make sure that there are no unclear parts
    - Solve the exercise, implement the algorithms yourselves
    - Take a note of any unclear parts/issues, and ask in the next lecture

[1] https://www.npr.org/sections/health-shots/2024/05/11/1250529661/handwriting-cursive-typing-schools-learning-brain



Common questions about CENG111

- Why do we “start” with Python
- Is Python the best/ultimate programming language?
- Will we learn Python to its full extent?
- Is it an advantage to “know programming” prior to 111?
- Shall I buy a desktop or a laptop? If yes, which brand and where to buy it?
- Shall I install Linux? Why are you so “UNIX” biased?

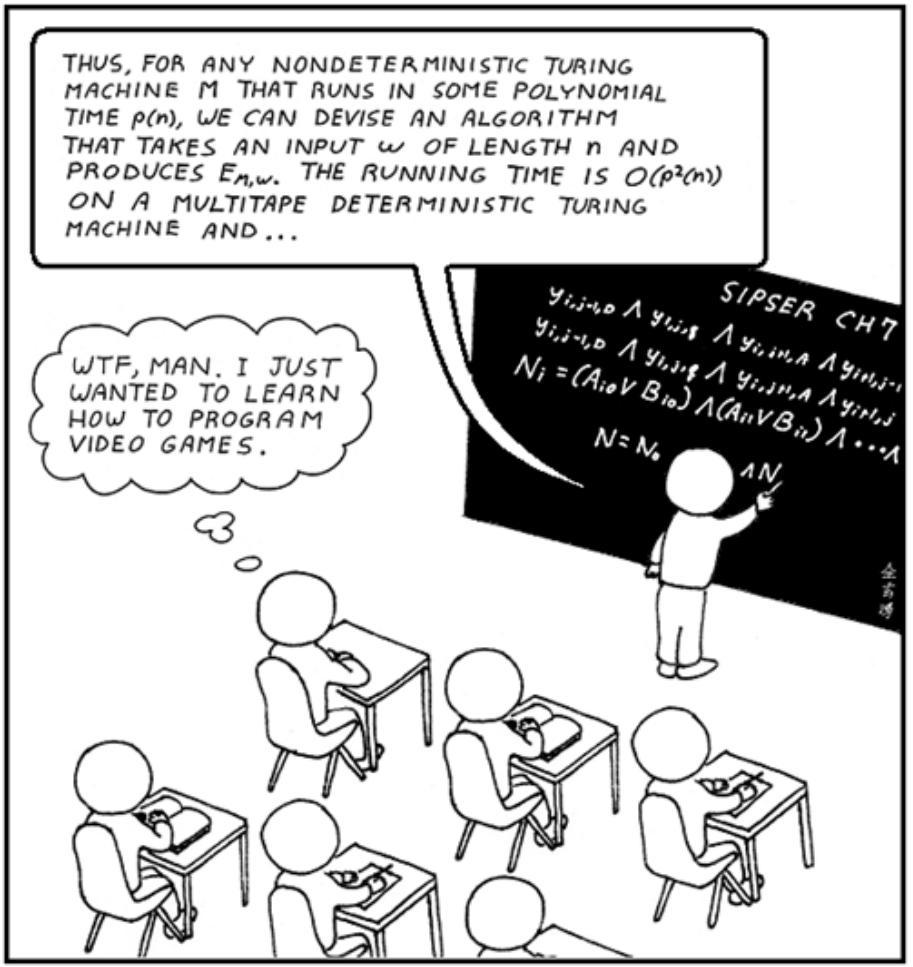
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Common questions about CENG111

- Can I take CENG 111 again next year?
  - No, unless you fail the course (FF/FD) or you are in probation

At the end

- CS is about mostly analysis, design, implementation, interpretation and usage of algorithms.
- If you are here for a different purpose, it is not too late ☺



At the end

- Although you might have some difficulties at the beginning, you will love it after 1-2 years
  - No guarantees though ☺

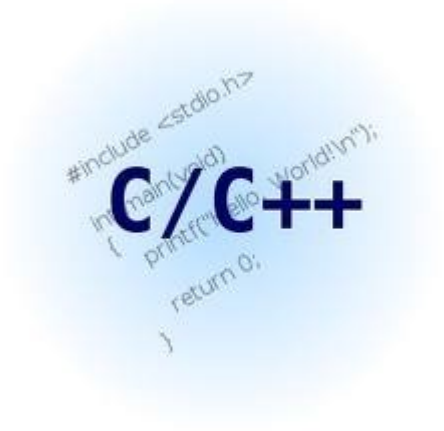


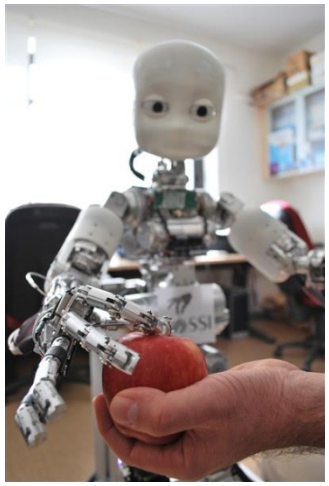




Now

- Introduction to Computer Science
  - What CS is and what it is not
  - Computing, computation, algorithm







WHAT IS COMPUTER SCIENCE?

What is Computer Science?

- *"Computer science is the study of computers"*

WRONG!

- Incomplete – theoretical work began (1920-1940) before computers
- CS became an independent field of study late 1950's, early 1960's
- Theoretical CS – relies on formal models rather than "real" machines
- CS "is no more about computers than astronomy is about telescopes"

From "Invitation to Computer Science"

What is Computer Science?

- *"Computer science is the study of how to write computer programs"*

WRONG!

- Programming is important, but it is just a tool for studying new ideas, representing information or testing the solution to a problem.
- A Program is a means to an end, not the end itself.

From "Invitation to Computer Science"

What is Computer Science?

- *"Computer science is the study of the uses and applications of computers and software"*

WRONG!

- E.g. word processors, databases, spreadsheets, etc.
- Many people USE software, but the Computer Scientist is responsible for specifying, designing, building and testing software packages and the systems on which they run.

From "Invitation to Computer Science"

What is Computer Science?

- All of the following concepts are incomplete and do not capture the richness and diversity of this exciting field:
  - computers,
  - programming languages,
  - software applications, and uses.

From "Invitation to Computer Science"



Computer Science is the study of algorithms (= methods)

including:

1. their formal and mathematical properties
2. their hardware realizations
3. their linguistic realizations
4. their applications

From "Invitation to Computer Science"



Computer Science is the study of Algorithms

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Including:

1. their formal and mathematical properties
   - studying the behavior of algorithms to see that they are correct and efficient

From "Invitation to Computer Science"

Computer Science is the study of Algorithms

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Including:

2. their hardware realizations
   - designing and building computer systems to execute the algorithms

From "Invitation to Computer Science"



Computer Science is the study of Algorithms

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Including:

3. their linguistic realizations
   - designing programming languages and translating the algorithms into these languages so that they can be executed by the hardware

From "Invitation to Computer Science"



Computer Science is the study of Algorithms

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Including:

4. their applications
   - identifying important problems and designing correct and efficient software packages to solve them.

From "Invitation to Computer Science"

Measure the height of a tall building with a barometer



- What would be your answer?
- One student answered:
  - "I would tie the barometer to a rope, hang it down from the top of the building to the bottom and measure the length of the rope!"
  - Of course, the instructor rejects the answer since it doesn't include any "physics"

Check the following for two different versions of the `legend':
http://www.snopes.com/college/exam/barometer.asp

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Measure the height of a tall building with a barometer (cont'd)

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- There are other ways:
  1. Drop the barometer from the top, and measure the time it takes to reach the ground.
  2. Make a pendulum and time its period wrt. the top and the bottom of the building.
  3. Walk down the stairs marking "barometer units" on the wall.
  4. Measure its shadow and the buildings shadow. Workout the height of the building from barometer's height.

Analyze/Compare Algorithms
= Pros and Cons of Algorithms =

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What are the disadvantages of these?

1. Drop the barometer from the top, and measure the time it takes to reach the ground.
2. Make a pendulum and time its period wrt. the top and the bottom of the building.
3. Walk down the stairs marking "barometer units" on the wall.
4. Measure its shadow and the buildings shadow. Workout the height of the building from barometer's height.

Cons:

1. You lose the barometer; it breaks.
2. You need a loooong rope and how long that rope is going to be depends on the answer to the question.
3. It takes too long. It is too tiring.
4. What if there is no sun?